



Observations show that stars form in large filamentary networks within Molecular Clouds (MC). The degree of fragmentation can vary significantly within the same cloud.

In this work, we investigate how gas dynamics regulate fragmentation and accretion in a colliding-flow simulation of the warm neutral medium.

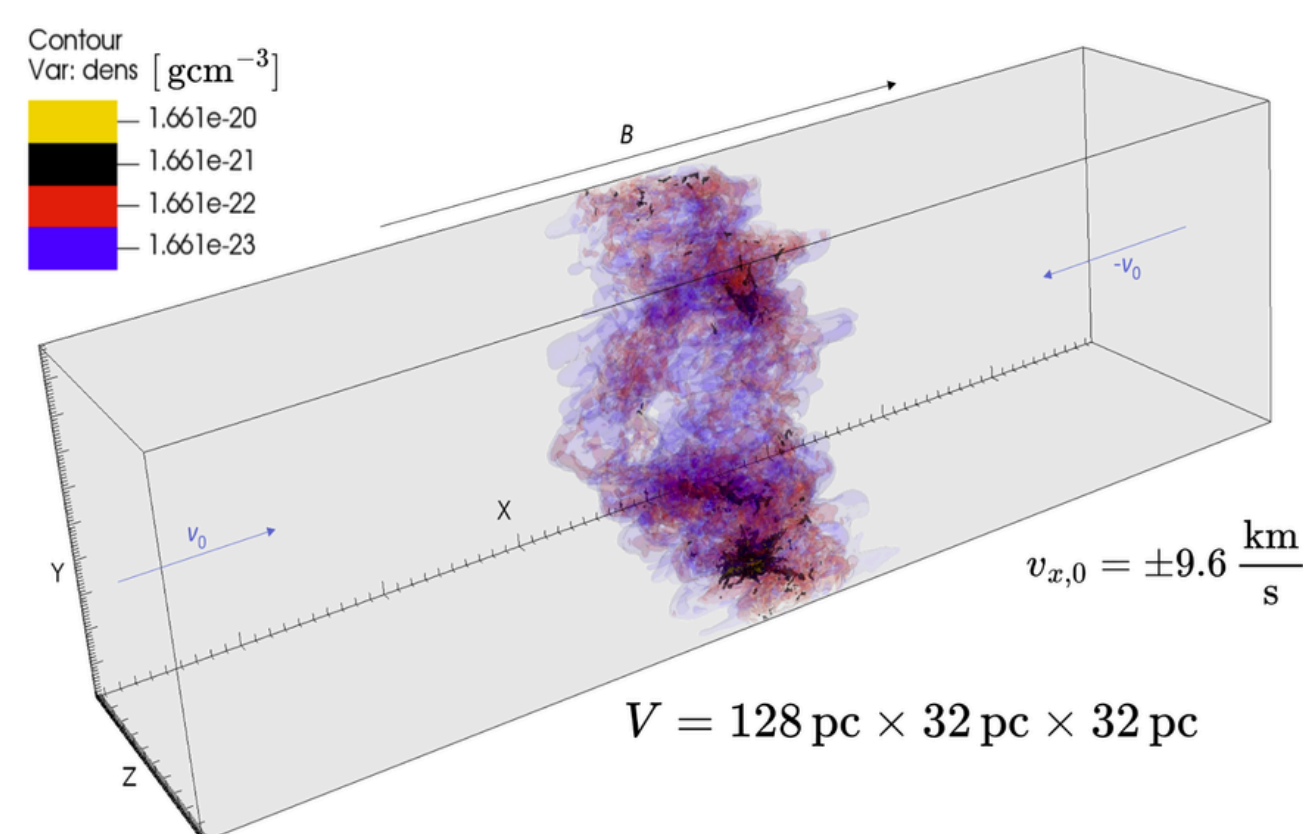


Figure 1: Sketch of the simulation setup by Weis et al. 2024.

Simulation: Setup

Colliding Flow Simulation by Weis et al.

- 3D MHD, adaptive mesh refinement (AMR) grid code **FLASH**
- Warm neutral medium inflow in the x-direction
- 2 million massless tracer particles (tracers) are initialized on a homogeneous grid and advected with the gas flow
- Maximum spatial resolution: 1600 AU
- Simulation time: 21 Myr

The two gas streams collide at an interface, and due to thermal instabilities, a dense sheet forms representative of a MC (Fig. 1). A simple chemical network is used to follow the formation of CO and H₂.

3D Clump & Core Tracking

Identification Algorithm by Weis et al. 2024:

- Filter AMR grid for cells with $\chi_{\text{CO}} > 1 \times 10^{-4}$
- Iterative neighbor-of-neighbor algorithm groups connected cells
- Clump found if at least 30 cells are connected
- For cores, an additional criterion: $A_{V,3D} > 8 \text{ mag}$
- Cores are shielded clumps: Dendrogram Analysis is applied on $\log_{10}(\rho)$

In Fig. 2, nodes represent either part of the core hierarchy or clumps with no substructure at each simulation snapshot (snap).

Each node is identified with a set of tracers. A merit score $\mathbf{m}$ is calculated for all nodes between two snapshots t and $t+1$, where:

$$\mathbf{m} = \frac{(|(\text{node } i, t) \cap (\text{node } j, t+1)|)^2}{|(\text{node } i, t)| \cdot |(\text{node } j, t+1)|}.$$

With this the output of the core and clump finder is represented by a traversable tracking graph.

The evolution of core leaves and clumps at the last snap is traced back using the Depth First Search (DFS) algorithm. DFS explores progenitor connections recursively until none remain, then backtracks to discover alternative pathways, referred to as lineages. We choose the main lineage as the path that maximizes the sum of merits.

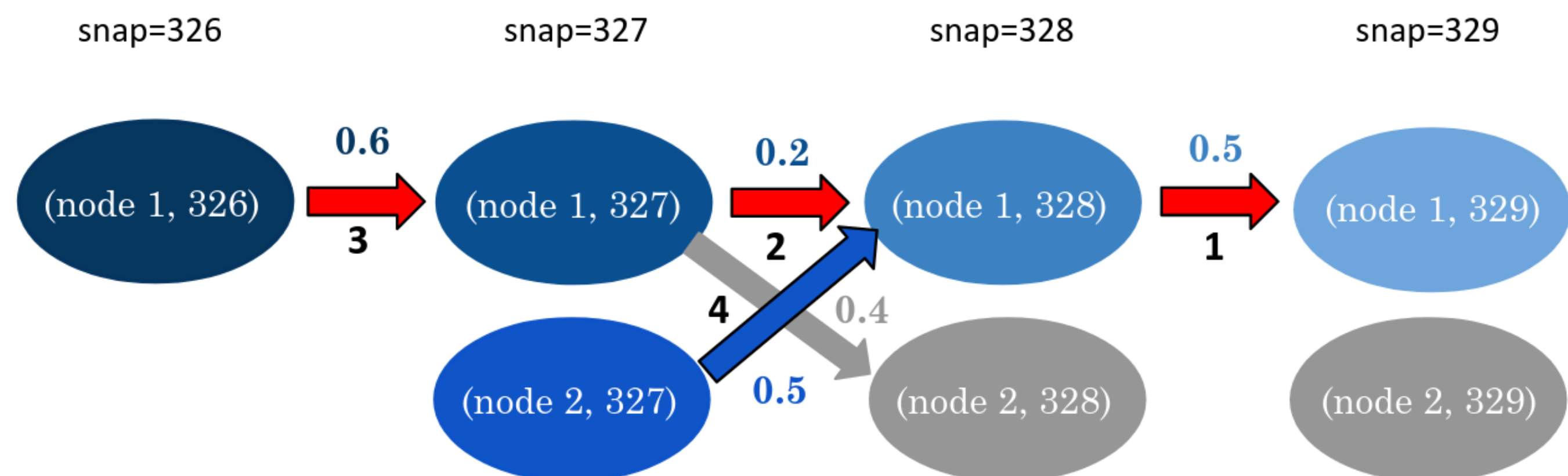


Figure 2: Illustration of DFS traversal of an exemplary tracking graph. The colored oval shapes represent nodes belonging to DFS lineages, while the corresponding colored numbers on the connecting arrows denote the merit score between the two nodes. The black numbers indicate the DFS traversal order. Grey nodes do not belong to any lineage, and (node 2, 329) does not satisfy the minimum merit threshold of 0.2. The red arrows indicate the main lineage of (node 1, 329).

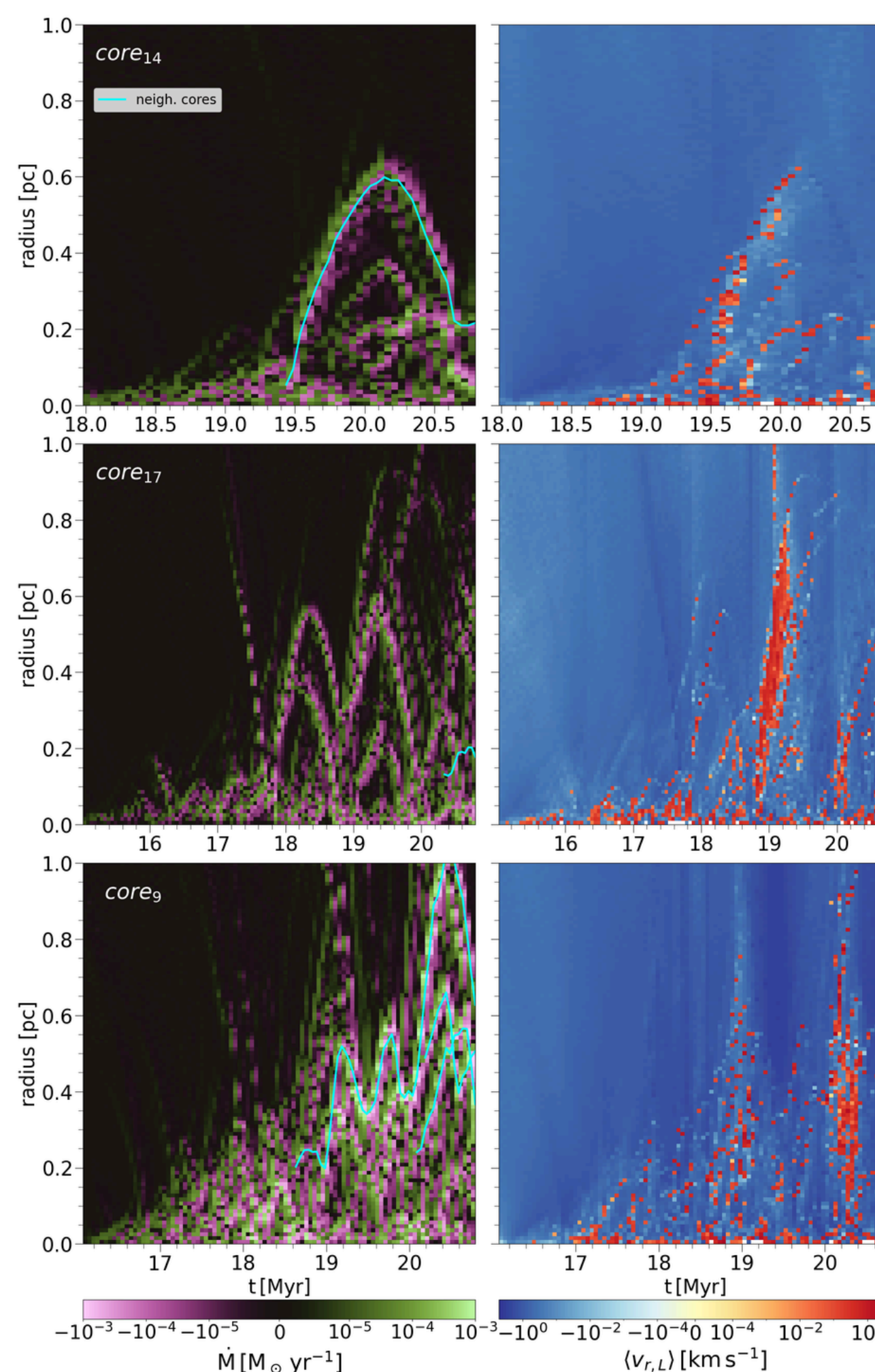


Figure 3: The first column shows the spatial and temporal evolution of the shell mass rate, while the second column shows the corresponding evolution of the shell-averaged radial tracer velocity. Each row corresponds to a different core used as the frame of reference. Negative radial velocities indicate flow towards the COM.

Global Mass Assembly

To analyse global mass assembly, we compute the shell mass rate and shell-averaged radial tracer velocity (Fig. 3). core₁₄ shows the simplest evolution, with most mass concentrated near the COM. core₁₇ is intermediate, with substantial mass oscillations around the COM but short-lived secondary cores. core₉ shows the most complex evolution, with mass extending to larger radii corresponding to long-lived secondary cores.

To investigate the influence of external gas dynamics on the central region, we compute the mass flux across spherical control volumes (Fig. 4). For early times the mass fluxes behave similar at both scales. At late times the mass flux varies due to mass crossing the boundary, especially on smaller scales. Thus, the central region becomes more active as mass is efficiently transported from larger to smaller scales. For core₁₇ the central region is more decoupled from the clump. For core₉ we see a smooth increase and decrease of mass flux between 19-19.8 Myr at large radii. At the same time a large scale inflow is prominent in the shell-averaged radial velocity and the central region becomes more dynamic suggesting a filamentary inflow.

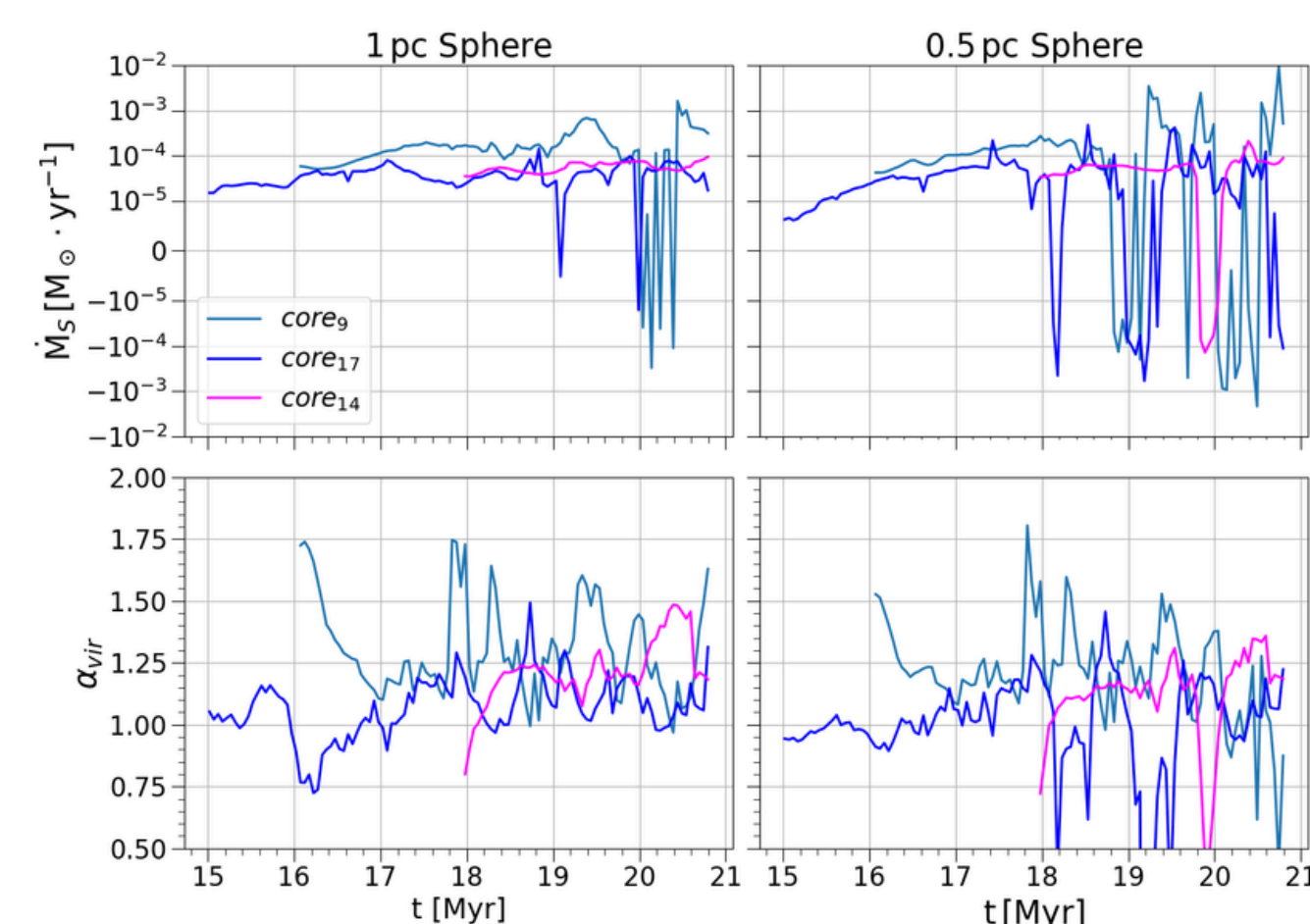


Figure 4: The upper panel shows the mass flux across a 1 pc and 0.5 pc spherical control volume centered on the core COM. The lower panel shows the virial parameter for both control volumes.

Eulerian Virial Analysis

For the same control volumes we compute the virial parameter, $\alpha_{\text{vir}} = \frac{E_{\text{HD}}}{|W|}$, $E_{\text{HD}} = 2(\epsilon_{\text{kin}} - \tau_{\text{kin}}) + 2(\epsilon_{\text{th}} - \tau_{\text{th}})$ where W is the gravitational binding Energy. For early times, core₉ is more kinetically and thermally dominated while core₁₇ is closer to equilibrium, especially on smaller scales. core₁₄ is almost scale independent. For intermediate times all cores behave similarly. At late times core₉ becomes gravity dominated at smaller scales, while the outer regions is dynamically active.

Parent clump	Core	$M_f [M_\odot]$
clump ₂ $M_f = 232.22 M_\odot$	core₁₄	90.45
	core ₂₁	29.92
clump ₀ $M_f = 408.02 M_\odot$	core₁₇	33.41
	core ₁₈	35.55
	core ₁₉	16.93
	core ₂₄	11.10
clump ₁ $M_f = 1115.13 M_\odot$	core₉	44.83
	core ₁₃	54.30
	core ₁₁	305.81
	core ₁₂	71.74
	core ₂₇	66.51
	core ₈	44.04

Table 1: First column shows the parent clump and final mass. Second column are all cores contained in the clump and the third column is the core final mass.

Summary & Conclusion

In total, we have three clumps and three scenarios (Tab. 1). clump₂ showed a scale independent evolution with one central massive core and one further fragment. core₁₇ evolved more decoupled from clump₀, with many short-lived fragments. clump₁ was the most strongly coupled to the central region, with multiple massive secondary cores.

Why do clumps within the same parent cloud exhibit different degrees of fragmentation?

The degree of fragmentation is dependent on clump-core coupling. The efficiency and amount of mass accreted is crucial for the further evolution.